AfyaPlus | Cost-Optimised AI Deployment

EXECUTIVE DECISION BRIEF | OCTOBER 2026 | USD PLANNING ESTIMATE

Recommendation

Keep a managed model API for the pilot and route eligible low-risk work to gpt-4o-mini. Preserve human review for high-risk or ambiguous cases. Defer GPU self-hosting at current modeled traffic.

Monthly cost scenarios

Scenario	Requests/month	Total / 1,000	Monthly total
Baseline	100,000	$5.3790	$537.90
Sustained 10x	1,000,000	$0.9790	$979.01

Token spend is $0.17901/1,000 at 450 input + 180 output tokens and a 2% retry allowance. Infrastructure and people/ops are disclosed planning allocations, not quotes. Verify current OpenRouter rates before budgeting.

Operating controls

Cache: 5/8 public FAQ fixture requests hit (62.5%); five-minute TTL; patient-specific answers are not cached. **Batch:** 7 mock offline jobs grouped into 2 local batches; no provider discount claimed.

Budget: proposed $1,000 monthly alert cap with 80%/100% actual and 100% forecast notices; alerts do not stop spend. Apply cost-center, environment, and workload tags. Bound autoscaling by quotas; use capped retries and a circuit breaker. Local Compose and written Azure/AWS configs are the fallback; no paid cloud, GPU, Redis, Kubernetes, or live load test is claimed.

Trade-off and decision gate

Managed API avoids fixed serving capacity and model operations, but retains provider price, quota, and availability exposure. The illustrative self-host break-even is about 15.25M requests/month ($1/hour GPU plus 40 operations hours/month); it is not a procurement result. Serverless cold starts and prompt/model A/B changes must meet p95 and quality gates. Quantization, pruning, and KV-cache tuning are self-host levers, not API savings. Validate current rates, throughput, retries, and quality at controlled 10x.